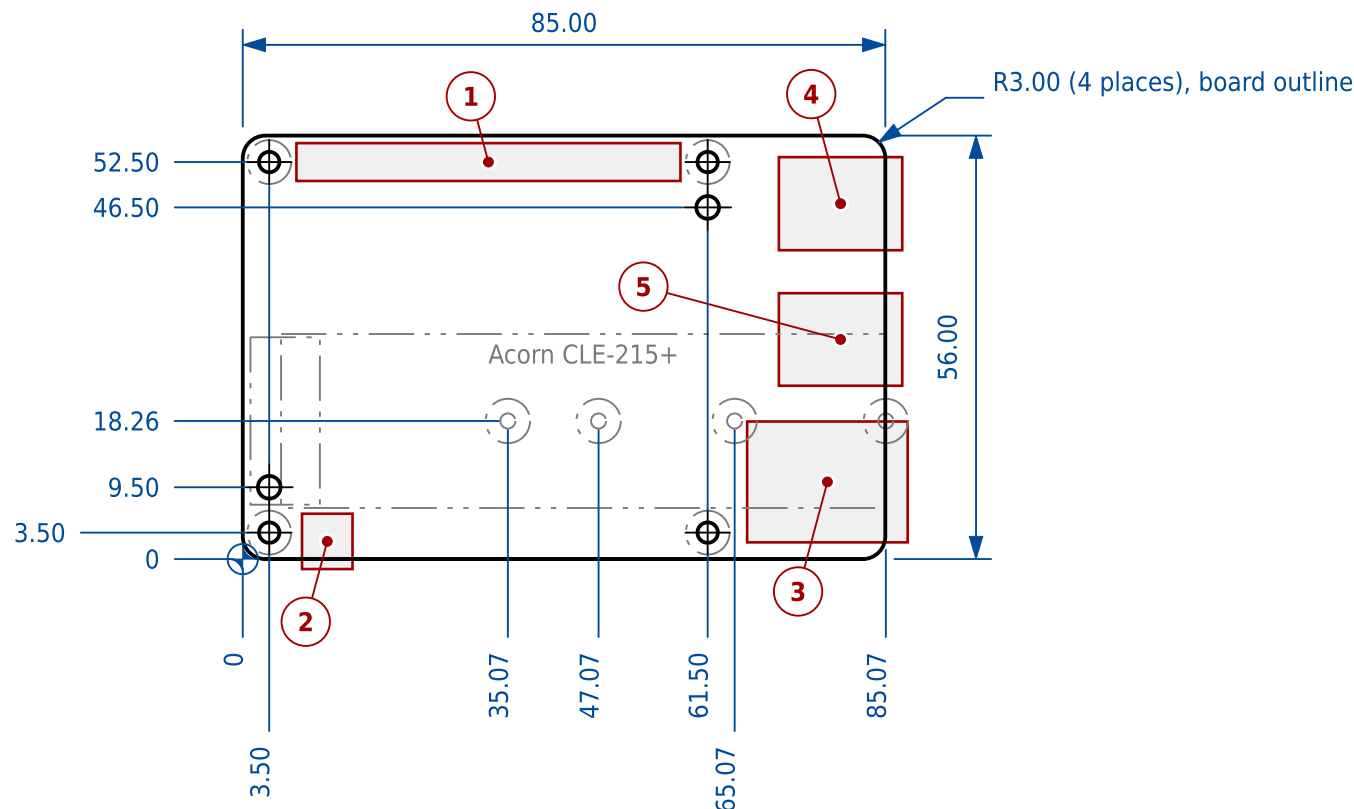




NOTES

1. Viewed from the component side.
2. Phantom outline is the Waveshare PoE M.2 HAT+ (B) fitted on the 40-pin GPIO header; its mounting holes coincide with this board's. Its own holes and bodies, and anything seated in them, are phantom too and are scheduled under its name.
3. The card's far end and its retention screw land at X 85.07, just past the Pi's Ethernet-end edge, and the standoff's 5.87 mm boss reaches 88.00: allow 3.00 mm past the 85 mm board, on the edge the RJ45 and USB already overhang.
4. The Acorn is 23 mm wide, a millimetre over the M.2 specification's 22 +/-0.15; SQRL say to make sure of the clearance.
5. PLAN ONLY, no Z anywhere: neither Waveshare nor SQRL publish a height for the HAT, its standoffs, the card, or the heatsink the CLE-215+ carries, which is therefore not drawn.
6. Waveshare dimension nothing of the M.2 system but the standoff's 3.00 mm overhang, so the socket, the standoffs and the card are DERIVED from their drawing by accessories/measure_poe_m2_hat.py, to +/-0.2 mm and the socket to +/-1 mm; the standoffs' DIA is the M2 x 0.4 tapped thread, not a clearance hole, and BOSS is what they occupy. MT1 to MT4 are not derived at all: their 2.75 mm is the Raspberry Pi HAT specification's DECLARED M2.5 clearance, which Waveshare do not dimension.
7. Assembled envelope, every overhang drawn included, the phantom part's as well: 88.00 x 57.32 mm.

SOURCES

1. Mechanical drawing: <https://datasheets.raspberrypi.com/rpi5/raspberry-pi-5-mechanical-drawing.pdf> - Raspberry Pi Ltd

SOURCES (continued)

2. Dimension drawing: <https://www.waveshare.com/w/upload/d/d9/PoE-M.2-HAT-Plus-B-details-size.jpg> - Annotated 85.00, 56.00, 58.00, 49.00, 3.50 and 3.00, "Unit: mm". The 3.00 is the 2280 standoff's projection past the board edge.
3. Product wiki: [https://www.waveshare.com/wiki/PoE_M.2_HAT%2B_\(B\)](https://www.waveshare.com/wiki/PoE_M.2_HAT%2B_(B)) - "Compatible with M.2 hard drives of 2230 / 2242 / 2260 / 2280 sizes", which of their three PoE M.2 HATs only this one is. Its "Product size: 56.5mm x 70.0mm" is the plain HAT+'s and contradicts this board's own drawing.
4. Raspberry Pi HAT mechanical specification: <https://github.com/raspberrypi/hats-hat-board-mechanical.pdf> - Waveshare dimension the hole CENTRES and not the holes: MT1 to MT4's 2.75 mm is this specification's declared M2.5 clearance. Its 58 x 49 mm rectangle 3.5 mm in from the edges is what their 58.00, 49.00 and 3.50 agree with.
5. PCI Express M.2 Specification: PCI-SIG, Revision 1.0, 1 November 2013 - Section 2.3.4.3 and figure 13: a Type 2280 module is 22 +/-0.15 by 80 +/-0.15 mm, its retention screw's half-moon cutout centred on the far end edge. Section 2.5.4.2 and figure 73: an M2 x 0.4 shouldered stand-off.
6. SQRL Acorn CLE-215+ product page: <https://web.archive.org/web/2020/http://www.squirrelsresearch.com/acorn-cle-215-plus/> - Squirrels Research Labs, captured 2020; the site is gone. "An M.2 2280 M-Key (PCIe) slot is required to use Acorn ... it is one millimeter wider than the official specifications. Ensure you have clearance."

HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	3.50	3.50	2.70	5.80
MT2	61.50	3.50	2.70	5.80
MT3	3.50	52.50	2.70	5.80
MT4	61.50	52.50	2.70	5.80
AUX1	3.50	9.50	3.00	not given
AUX2	61.50	46.50	3.00	not given

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	40-pin GPIO header	7.10 to 57.90	50.00 to 55.00
2	USB-C power input	7.84 to 14.52	-1.32 to 6.00
3	Ethernet RJ45	66.72 to 87.96	2.21 to 18.19
4	USB 3.0 type A (upper pair)	70.92 to 87.24	40.84 to 53.15
5	USB 3.0 type A (lower pair)	70.92 to 87.24	22.92 to 35.16

WAVESHARE POE M.2 HAT+ (B): HOLES

ID	X mm	Y mm	DIA mm	BOSS mm
MT1	3.50	3.50	2.75	not given
MT2	61.50	3.50	2.75	not given
MT3	3.50	52.50	2.75	not given
MT4	61.50	52.50	2.75	not given
2230	35.07	18.26	2.00	5.87
2242	47.07	18.26	2.00	5.87
2260	65.07	18.26	2.00	5.87
2280	85.07	18.26	2.00	5.87

WAVESHARE POE M.2 HAT+ (B): BODIES

ITEM	X EXTENT mm	Y EXTENT mm
M.2 M-key socket	1.02 to 10.20	7.19 to 29.33
SQRL Acorn CLE-215+, M.2 2280 M-key	5.07 to 85.07	6.76 to 29.76

LEGEND

- Board outline
- Component body, scheduled
- Adjacent part, hole keep-out
- Dimension, extension and leader

ACORN CLE-215+ IN A POE M.2 HAT+ ON A PI 5

Plan envelope of the assembly; the Raspberry Pi 5 is the board drawn

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-445-g91dd743	mm	1:1
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	ACC-HAT-M2POE	A
MATERIAL	GENERAL TOLERANCE		
PCB, thickness not stated	edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08		